

Schedule A-15

This Statement of Work, (hereafter, “**SOW**”) Number A-15, dated January 02, 2024, is issued pursuant to, and is hereby incorporated by reference in, that certain Master Services Agreement dated as of April 29, 2020, and as amended, modified or supplemented from time to time (the “**Agreement**”) between NVISIA (“**NVISIA**” or “**Vendor**”) and Federal Home Loan Bank of Chicago (the “**Bank**”).

All capitalized terms not otherwise defined herein shall have the meaning assigned to such term in the Agreement. In the event of any inconsistency, the terms or conditions of the Agreement shall control, except that a term in this SOW that specifically references a conflicting term or condition of the Agreement will control solely with respect to the Services covered under this SOW.

1. **GENERAL DESCRIPTION OF SERVICES**

Within the MCS department, the Member Experience Team (MET) is responsible for the development and support of the multiple applications including eBanking, CSAP, MTAP, Direct Wire and Member API. Based on the roadmap for the MET team, the NVISIA resources will be working on all MET applications to support Bank priorities. A few approved projects include, but are not limited to, EFT Gold, Salesforce 2.0, Data Lake House, or CAMP 3.0. The delivery process manager (DPM) will be responsible for the entire MET team.

The eBanking system serves as both an internal application and a business to business (B2B) portal where FHLBC members can access account information, execute both advances and letters of credit, extract data, run reports, as well as a handful of many other functions. When updating eBanking the goal is to redesign and replatform the current java based eBanking application to Angular/.NET Core/SQL Server while also adding multiple features and functions via a fully responsive web interface to improve the user experience for internal staff and our members. The application must scale and load balance dynamically, benefit from a micro services architecture, leverage AWS, use the Banks suite of DevSecOps tooling and processes, integrate with multiple core Banking systems via REST API's while also leveraging automation and robust monitoring solutions. Both security and quality will be paramount, as business transactions are typically low in volume but high in value.

2. **DESCRIPTION OF DELIVERABLES**

During the duration of this SOW, the Bank and NVISIA will work together to set priorities, adjust sequencing of development activities via the agile roadmap, and implement stated functionality within the Bank through an agile, iterative process.

The Bank will be leading the efforts as NVISIA will support the MET team in a staff augmentation model by performing the following:

- Help deprecate existing functionality and migrate approved features to new application
- Leveraging the Bank's Checkmarx tool, ensure no vulnerabilities (no critical, high, medium or low) move into QA or production regions when inspected against the OWASP Top 10 and Sans 25 databases

- The new application must be fully responsive to support the following form factors (PC, tablet and phone) in addition to fully functioning on the current versions of both Google Chrome and Microsoft Edge web browsers
- The tooling and tech-stack versions need to stay current and will be upgraded over the life of the project with such future upgrades and the timing thereof being determined and approved by the Bank
- At least one code review should be performed by a Bank architect or Bank assigned delegate
- Adhere to all Modern Engineering pillars of people, process and technologies for pipelines, provisioning and configuration of all server builds and test case repositories
- The amount of test regions will be updated in the test plan and executed through IaC
- Tracking of Bank dependencies will be the responsibility of the Delivery Manager to ensure the project stays on track. All roadblocks are identified and the Delivery Manager will coordinate the resolution of each
- UI/UX – if required, prototypes will be created or modified for features identified in the project. Low fidelity mockups will be created for all main screens identified in the site map. High fidelity mockups will be created for features which introduce new UI components in order to support development efforts.
- All transition activities will be identified and executed to ensure the Bank staff can support the application at the end of this engagement

Being an agile effort, specifics of these expectations may change or be adjusted as the project progresses. Further, the project will be run an agile framework and the deliverables will be prioritized by the product owner and may change.

All deliverables identified in Sections 2, above, include the delivery of all relevant documentation, including requirements/architecture/design specifications.

3. BANK RESPONSIBILITIES

The Bank shall perform the tasks, functions and responsibilities as set forth below:

- Designate the Bank individuals who possess the suitable skills, knowledge and/or experience to support this project and serve as the Project Sponsor, Product Owner and Stakeholders with relevant source systems knowledge as well as financial industry domain knowledge.
- Provide access to, and make available, the Project Sponsors, Product Owner, Stakeholders and other agreed key personnel as reasonably required during the course of the project.
- Provide, on a timely basis (generally within one to three business days), such information, documentation, and assistance that are necessary to NVISIA's performance of the Services or that NVISIA reasonably requests.
- Provide an IT infrastructure access (including systems and application environment access through Bank provided VPN systems as reasonably necessary to perform the Services described within this SOW, subject to the requirements set forth in Schedule B-1 of the Agreement, "FHLBC IT Use and Security Policy."

- Use reasonable commercial efforts to provide to NVISIA complete, accurate, current and relevant information during the course of the project.
- The Bank is responsible for the quality of data provided to NVISA that is sourced from non-eBanking business applications. Any activities such as data cleansing, de-duplication, or de-identification or changes needed in other source system data are to be performed with Bank resources.
- Provide at least one development environment and one test environment.
- Ensure use and procurement of appropriate licenses.
- Produce all regulatory, business rule, subject matter and other requirements documentation specific to eBanking as necessary.
- The Bank is responsible for specifying and providing security or penetration testing tools together with necessary penetration testing.

Leadership / Project Sponsors / DPM

- When possible, attends key Agile ceremonies
- Reviews the proposed release plan for the system
- Reviews and provides input into Planning Deliverables
- Participates in weekly status review meetings
- Helps in removal of roadblocks when unable to be removed by the team

Solutions Architect

- Provides context into current state
- Provides NVISIA with Bank policies, standards, procedures and guidelines in alignment with the delivery life cycle (DLC) and the software development life cycle (SDLC)
- Partners with the NVISIA team to ensure proposed architecture and tooling is in alignment with FHLBC enterprise architecture standards and aligns with the Modern Engineering (DevSecOps) vision for the Bank
- Reviews and provides input into Phase1 Architecture Deliverables
- Approves all user stories that are considered to be “technical user stories” prior to pulling them into a sprint
- Attends Agile team ceremonies

Product Owner

- Shares and provides context into overarching product vision and strategy
- Works with NVISIA team to identify, prioritize and approve Epics, Features and User Stories
- Actively participant in discovery sessions
- Reviews and provides input into Phase1 Product and UI/UX Deliverables
- Coordinates with stakeholders to a review of the proposed prototype and helps to gather appropriate feedback

- Attends daily stand ups, planning, grooming, demo ceremonies in addition to any working sessions
- Approves all user stories that are considered to be “business functionality” prior to pulling them into a sprint

SME's / Stakeholders

- Provides input to the product vision
- Provides requirements for the system
- Actively participates during shadowing and discovery sessions
- Reviews and provides input into the UI/UX Deliverables
- Attends product demos when appropriate

Developers / QA Tester

- Reviews and provides input into design patterns and testing plan/strategy
- Writes application code or scripts in support of the system build and testing efforts
- Reviews and provides input into a formal transition plan
- Provides input into the DevSecOps tooling and approach
- Attends Agile team ceremonies

4. KEY VENDOR POSITIONS

Role Titles may include some of the following roles:

1. NVISIA Director
2. Delivery Manager aka Delivery Process Manager
3. Solution Architect
4. Technical Architect
5. Technical Lead
6. Software Engineer
7. Software Developer
8. Lead Business System Analyst
9. QA Manager
10. Lead QA Engineer
11. QA Automation Engineer
12. UX Product Architect
13. Product Design Architect
14. DevOps Engineer

Role Title: NVISIA Director

Responsibilities: Delivery Oversight, Leadership Communication

Role Title: Delivery Manager aka Delivery Process Manager

Responsibilities: Overall Delivery Leadership, Dependency Management, Visibility into Delivery Progress, Status Reporting, Support Product and Creative.

Role Title: Solution Architect

Responsibilities: Enterprise Architecture, Security, Application Architecture, and foundational components required for development.

Role Title: Technical Architect

Responsibilities: Partners with Solution Architect to formulate application designs and ensure their alignment with enterprise guidance and standards; provides daily technical leadership to implementation team while contributing to solution implementation; develops Proof of Concept (POC) applications at the direction of and in collaboration with the solution architect; provides technical input for the creation and refinement of product requirements; assists Solution Architect in documenting processes, procedures, and standards; assists in developing test automation scripts as needed.

Role Title: Technical Lead

Responsibilities: Assists in validating architecture and flushing out DevSecOps tooling and approach

Role Title: Software Engineer/Developer

Responsibilities: Assists in developing application and infrastructure.

Role Title: Lead Business System Analyst

Responsibilities: Product Roadmap creation, product requirements elicitation and definition via epics, features and stories

Role Title: QA Manager

Responsibilities: Develop Test Framework, Refine QA Process, Establish QA Criteria

Role Title: Senior QA Engineer

Responsibilities: Develop Test Framework, Refine QA Process, Establish QA Criteria

Role Title: UX Product Architect

Responsibilities: Initial User Research, UI Design and Documentation, User Feedback Sessions

Role Title: Product Design Architect

Responsibilities: Initial User Research, UI Design and Documentation, User Feedback Sessions

Role Title: DevOps Engineer

Responsibilities: Creates DevSecOps strategy, tooling and approach and ensures alignment with the Bank's Modern Engineering initiative

Additional NVISIA Responsibilities

- Review and follow applicable Bank Policies, Standards, Procedures and Guidelines provided to NVISIA.
- Ensure the team is working in/adhering to the Banks delivery life-cycle (DLC) and software development life-cycle (SDLC) methodologies
- Responsible for leading all agile ceremonies.

- Ensure the usage of AWS / Modern Engineering related technologies upon approval from the Bank's architect. Perform all development from the Bank's AWS account.
- Review the Bank's tool set and provide recommendations to the Bank when NVISIA sees opportunities for change and/or improvement.
- Ensure that deployments adhere to the Bank's Change Management process. When appropriate, attend Change Advisory Board (CAB) meetings to represent upcoming changes.
- Present proposed DevSecOps tooling and approach to the Modern Engineering CoE team to ensure alignment with the Bank's Modern Engineering vision.
- NVISIA will utilize reasonable efforts to retain and maintain assigned resources throughout this project. In an unforeseeable event such as voluntary resignation, involuntary termination for cause, illness, disability or death of any of the assigned resources during the term of this SOW, NVISIA shall (i) give Bank as much notice as reasonably possible of this development, and (ii) will expediently identify and obtain a replacement resource with comparable knowledge and training to complete the Services with at least the same degree of skill and efficiency as the previous resource. In any event, the Bank will not be obligated to pay for any hours of training or knowledge transfer between the departing resource and the replacement resource. The Bank has the right to remove NVISIA resources from the NVISIA team at its discretion. NVISIA will use commercially reasonable efforts to provide an acceptable replacement within three (3) business days. NVISIA may not add resources to the project unless approved in writing by the Bank. The addition of resources shall not increase the maximum cost set forth in Section 11 of this SOW.

5. AUTHORIZED SUBCONTRACTORS AND ASSIGNMENT

The Vendor may not directly or indirectly engage subcontractors to fulfill obligations under this SOW unless (i) subcontractor is identified in the SOW, or (ii) Vendor otherwise obtains written approval from the Bank. Additionally, in accordance with Section 12.2 of the Agreement, Vendor may not assign, sell or otherwise dispose of this SOW, in whole or in part, without the prior written consent of the Bank.

The following subcontractors are identified and accepted by the Bank for this SOW: none

6. SCHEDULE

The delivery team will follow a SCRUM model for delivery. NVISIA and FHLBC will collaborate on a release plan which will contain multiple production deployments over the course of the engagement.

7. TESTING PROCEDURES AND ACCEPTANCE CRITERIA

The development team will test all the software in accordance to the testing strategy, including functional testing, unit testing, integration testing, code quality and security testing.

The following testing procedures will be followed:

- Develop and maintain automated functional test scripts for the completed user stories. All tests will be created and maintained at the unit, API and UI levels as appropriate to achieve full automation test coverage of the application
- Execute testing including system, integration, performance and load testing.
- Test user stories in accordance to the Definition of Done
- Execute regression testing as defect fixes are verified to validate that previously working features have not been adversely impacted by recent fixes
- Business stakeholders will execute user acceptance testing and provide written signoff upon completion

8. LOCATION

Vendor will provide the Services and deliver final deliverables to the following location(s):

Federal Home Loan Bank of Chicago, via remote connection provided by the Bank or at the Bank's Office, 433 West Van Buren Street, Chicago, or NVISIA Office, 200 South Wacker Drive, Chicago, as approved by the Bank. Provided however that given current COVID-19 restrictions Bank and Vendor agree that the Services are currently contemplated to be delivered remotely. The parties will consult from time to time as to any changes or adjustments to the Services delivery method.

9. SERVICE LEVELS

Vendor will perform the Services in accordance with the Service Levels set forth in this Section 9. If Vendor fails to meet the applicable Service Levels, Vendor will award Bank Service Credits as set forth below.

N/A

10. TERM

The Term of this SOW shall begin on January 02, 2024 ("**Effective Date**") and shall expire December 31, 2024 or until the Deliverables are accepted by the Bank as provided by the Agreement, whichever occurs first, unless the parties mutually agree to extend the SOW by email at least 10 days prior to the expiration date, in which case, the term shall expire on the date to which this SOW is extended.

Notwithstanding the foregoing, either party may terminate this SOW pursuant to the terms of Section 3 of the Agreement.

11. CHARGES AND BILLING

Fees and Expenses:

The Services will be billed on a time and materials basis in an amount not to exceed \$1,908,709 (\$1,461,509 in capital and \$447,200 in expense). Fees for Services provided by NVISIA will be billed

based on the number of hours worked (not to exceed 40 hrs. a week unless approved in writing by the Bank) and shall be charged at the 2024 nvisia rates that were agreed upon by FHLBC in 2023. The services will be provided through December 31, 2024.

The hourly rates and discounts are reflected in the charts below:

Role	Rates
Delivery Manager – Carter Cates	\$ 210
Technical Architect – Filip Radzikowski	\$ 185
Software Engineer – Trenton Daniels	\$ 145
Lead Business System Analyst – Megan Heller	\$ 165
Technical Lead – John-Paul Ensign	\$ 170

Volume Discount Schedule				
Level	Spend	Spend at Level	Level Discount %	Level Discount Amount
Level 1	\$ 0 – \$ 1,000,000	\$1,000,000	1%	\$10,000
Level 2	\$ 1,000,001 – \$ 1,500,000	\$500,000	2%	\$10,000
Level 3	\$ 1,500,001 – \$ 2,000,000	\$500,000	3%	\$15,000
Level 4	\$ 2,000,001 and over	Unlimited	4%	Unlimited

Invoices are to be submitted to the Bank's Accounts Payable department by NVISIA, as soon as practicable after the first business day of the calendar month, for the previous calendar month's work in the form of progress payments that will represent the value of the work completed and with the Bank to designate the invoice approver. Project time will be documented by the appropriate NVISIA resource in the NVISIA time entry tool. Hours spent will be reported to the Bank on a weekly basis, and reflected on each itemized monthly invoice. Bank shall pay all undisputed invoices within thirty (30) days of receipt.

Vendor will be responsible for all expenses incurred by the Vendor in the performance of the Services. No travel is authorized under this SOW. If the Bank directs or if NVISIA determines that daily travel is required to locations greater than 100 miles from the Bank's location, or if any overnight stay at a location greater than 100 miles from the Bank's location is required, NVISIA will submit a Change Order to this SOW for qualified travel costs.

12. BANK SOFTWARE AND BANK MATERIALS

Subject to the requirements of Schedule B-1 of the Agreement, Vendor shall have access to the following Bank network elements and Bank materials in performing the Services under this SOW:

- Bank will provide access to systems and tools required to perform the work stated within this SOW.
- Bank will provide documentation around current state eBanking system.

Subject to the requirements of Schedule B-2 of the Agreement, do the Services covered under this SOW require the Vendor to maintain or store Bank Customer Data as defined in Section 5.1 of the Agreement outside of the Bank's network?

Yes – ☐

No – ☒

If the answer above is "Yes", the vendor shall comply with requirements of B-2 attached to the MSA.

13. ADEQUACY OF SERVICES AND RELIANCE

Vendor represents and warrants that the Services and Deliverables set forth in this SOW are adequate to complete the portion of the project as to which the Vendor is responsible and compliant with applicable industry standards. Additionally, Vendor acknowledges and agrees that the Bank has relied on Vendor's expertise in determining the appropriate Services and Deliverables to complete the project and the scope of this SOW, with completion subject to the Bank's Responsibilities identified herein.

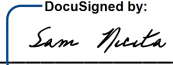
14. TRANSITION SERVICES

If requested by the Bank and subject to written agreement, the Vendor shall provide transition services at its then-current rate to assist the Bank in transitioning Services to either the Bank or its designated third party as defined in the Agreement.

15. APPROVAL

The Parties hereto have caused this Statement of Work to be executed by their respective duly authorized representatives as of the Effective Date hereof. All other terms and conditions of the Agreement not expressly modified herein remain unchanged and in full force and effect.

FEDERAL HOME LOAN BANK OF CHICAGO

Signature: 
DocuSigned by:
CC655F621E47487...
 Print Name: Sam Nicita
 Title: EVP/CIO
 Date: 12/14/2023

NVISIA

Signature: 
DocuSigned by:
B8BD0F74D5C64BB...
 Print Name: Shaun Lovick
 Title: President
 Date: 12/14/2023